Arithmetic Operation of Signed & Unsigned Numbers

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1's Complement

- In simple words, 1's complement is exchanging all the 0's into 1 and all the 1's into 0 of any number.
- Actuality, the one's complement means the addition of a negative integer to the number, and this eliminates the requirement of a separate subtraction processor.

$$\begin{array}{cccccccc} 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \end{array}$$

1's Complement

Application of 1's Complement:

- Represent Signed binary number: 1's complement is used to represent signed binary numbers, as positive signed binary numbers can be represented as it is, 1's complement is widely used to represent negative binary numbers in sign-magnitude.
- 1's complement is also used in various arithmetic binary operations like subtraction, addition, etc.

1's Complement

Question 1: $(110101)_2 - (100101)_2$

Solution:

$(1\ 1\ 0\ 1\ 0\ 1)_2 = 53_{10}$

$(1\ 0\ 0\ 1\ 0\ 1)_2 = 37_{10}$ – subtrahend

Now take the 1's complement of the subtrahend and add with minuend.

1 carry

```
        1 1 0 1 0 1
   (+)  0 1 1 0 1 0
        ---------------
        0 0 1 1 1 1
        1 carry
        ---------------
        0 1 0 0 0 0
```

Therefore, the solution is 010000

$(010000)_2 = 16_{10}$

1's Complement

- **Question 2: $(101011)_2 - (111001)_2$**

Solution:
Take 1's complement of the subtrahend

```
1 1 1
        1 0 1 0 1 1
   (+)  0 0 0 1 1 0 (1's complement)
        ------------
        1 1 0 0 0 1
```

Now take the 1's complement of the resultant since it does not carry 1
The resultant becomes 0 0 1 1 1 0

Now, add the negative sign to the resultant value
Therefore, the solution is $-(001110)_2$.

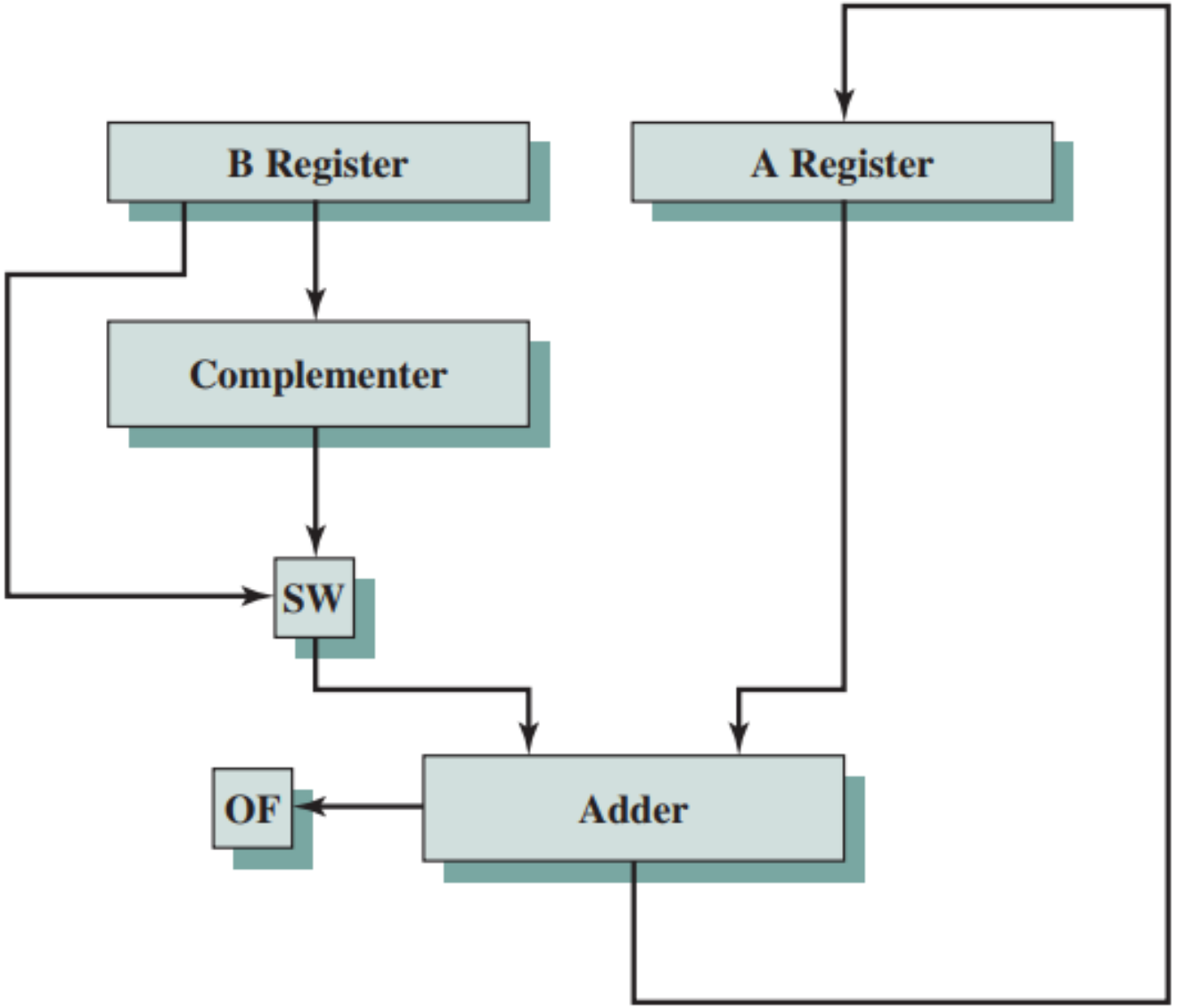
2's Complement

- Represents both positive and negative numbers.
- To compute 2's complement we need to add 1 to its 1's complement.
- 2's complement = 1's complement +1
- Represent -6 using 2's complement: we need to do 1's complement of +6 and add 1 to it. +6: 0110 1's complement of +6: 1001 then 2's complement=(1001+1)=1010, which is the required representation.
- +13: 01101
- -13: (10010+1)=10011
- The range of signed magnitude is – 2 n-1 to (2n-1 -1)

Block Diagram of Hardware for Addition and Subtraction

Step	Addition	Subtraction
1	Load A and B	Load A and B
2	Pass values to the adder	Convert B to two's complement
3	Compute sum	Compute A + (−B)
4	Store result	Store result
5	Check overflow flag	Check overflow flag

Block Diagram of Hardware for Addition and Subtraction



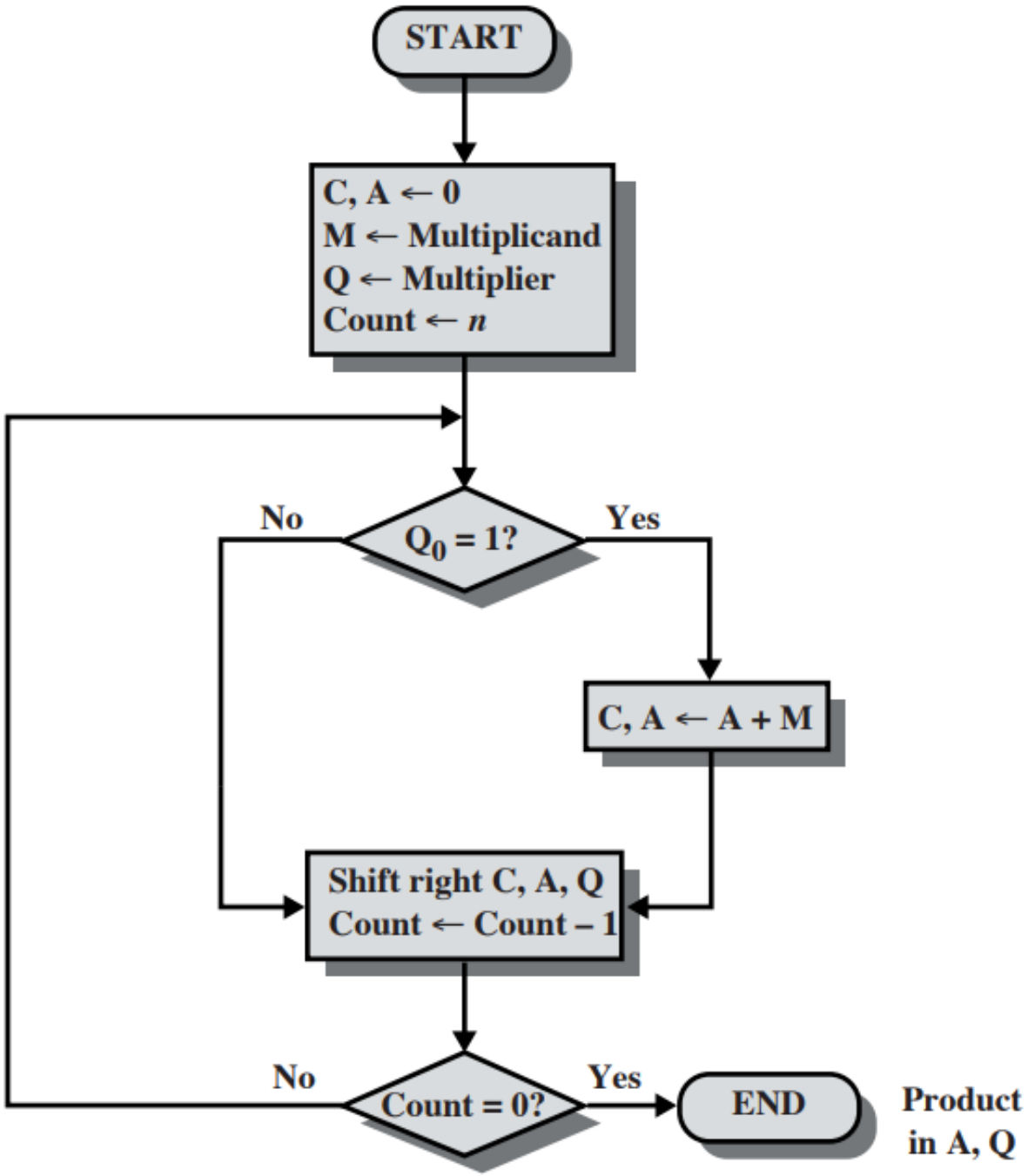
OF = Overflow bit
SW = Switch (select addition or subtraction)

Multiplication

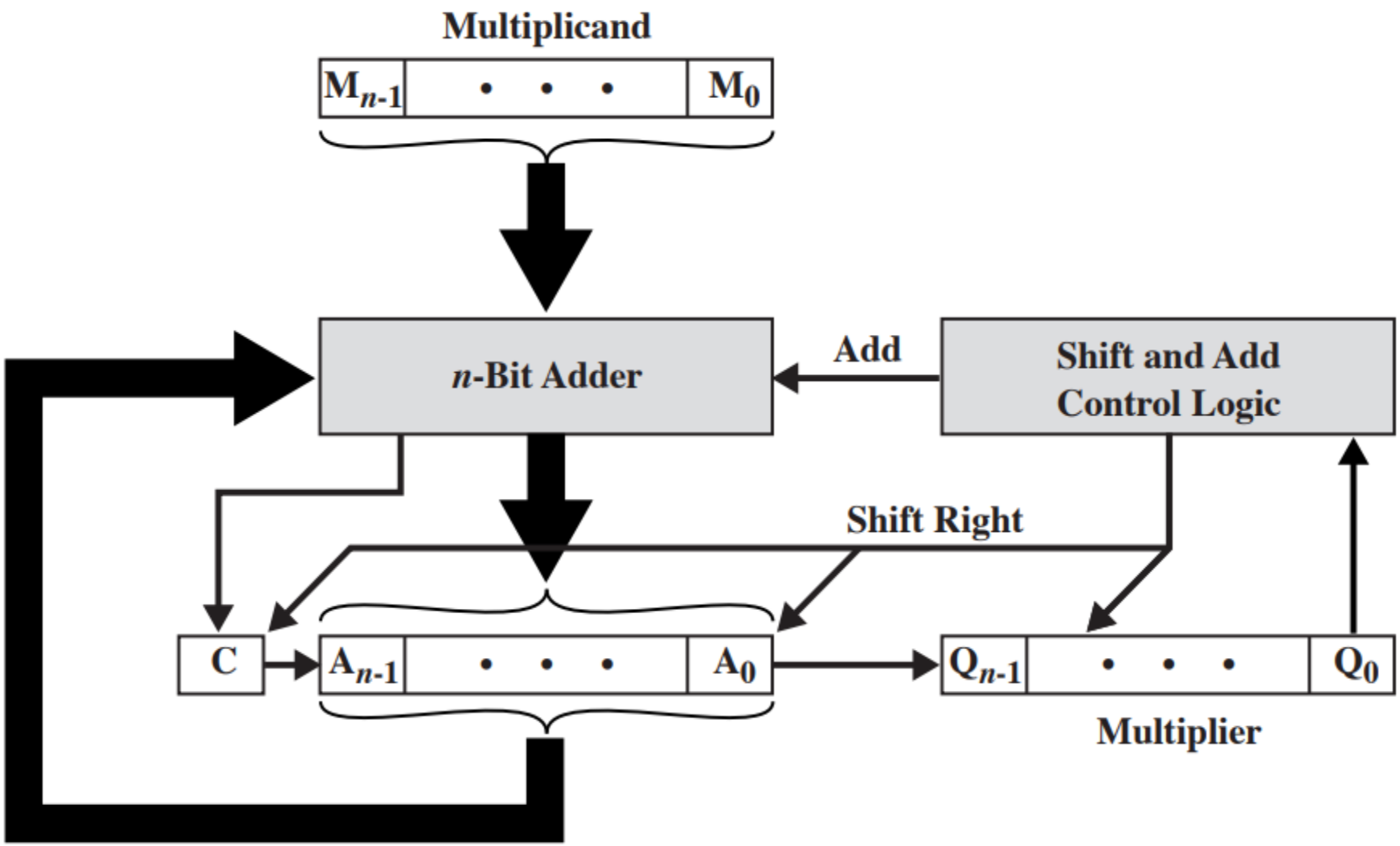
- **Unsigned Multiplication:**
  - Unsigned multiplication is a multiplication operation where both numbers are treated as positive (non-negative) integers.
  - It follows the basic rules of binary multiplication without considering any sign bit.

- **Signed Multiplication:**
  - Signed multiplication involves numbers that can be positive or negative.
  - The multiplication follows the same binary multiplication rules, but special attention is given to the sign of the numbers.

Unsigned Binary Multiplication



Unsigned Binary Multiplication

C	A	Q	M		
0	0000	1101	1011	Initial Values	
0	1011	1101	1011	Add	First Cycle
0	0101	1110	1011	Shift	
0	0010	1111	1011	Shift	Second Cycle
0	1101	1111	1011	Add	Third Cycle
0	0110	1111	1011	Shift	
1	0001	1111	1011	Add	Fourth Cycle
0	1000	1111	1011	Shift	

Hardware Implementation of Unsigned Binary Multiplication